(1.) Course Title: INFPHD601 Design and analysis of algorithms (complex exam preparatory)	**Credits: 6**
Course **Classification:** mandatory chosen	
Theoretical or Practical Nature of the Course, *'training character'*[13]: 80% theory, 20% practice (credit%)	
Type of Class: lecture and **number of hours**: **2/week** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: -	
Assessment Method (Exam / Midterm Grade / **Other**[3]): colloquium **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: -	
Course Placement in Curriculum (Semester Number): 4. semester (autumn)	
Prerequisites (*if any*): -	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

- Basic computational models (recursive functions, Turing machines, RAM-machines), their equivalence. Algorithmically not solvable problems.
- Fundamentals of analysis of computer algorithms. Time and space complexity, asymptotical behaviour of functions, recurrences and three methods for solving recurrences (substitution method, recursion-tree method, master-method), probabilistic analysis.
- Most important classes of problems form practical point of view: P, NP, NPC. The most important NP complete problems.
- Sorting problem. Comparison sorts and the theoretical lower bounds for the number of comparisons. The most important comparison sorts (bubble sort, insertion sort, heap sort, quick sort, merging sort, sort trees) and their analysis.
- Sorting in linear time, counting sort, radix sort, bucket sort.
- External sorting. Two-way merging with three buffers, balanced and Fibonacci versions. General extended merge sort. Method for growing the lengths of runs, replacement selection for growing the lengths of runs
- Sorting networks.
- Selection problem. Maximum, minimum, simultaneous maximum and minimum, median and order statistic. Theoretical lower bounds for comparisons. Selection in expected linear time and worst-case linear time.
- Searching problem. Hashing (direct address tables, hash tables, collision resolution by chaining, open addressing, construction of hash functions).
- Searching problem. Linear and binary searching. Searching trees (basic version, optimal searching tree, AVL-trees, red-black trees, 2-3 trees, B-trees, splay-trees, tries).
- Operations on strings. String-matching algorithms (brute-force, Boyer-Moore, RabinKarp, Dömölky-filter, string-matching automata). Longest common subsequence
- Elementary graph-algorithms (breadth-first search, depth-first search, topological sort, strongly connected components).
- Advanced graph-algorithms (minimal spanning trees, single-source shortest paths, allpairs shortest paths).

Geometrical algorithms (algorithms on line segments, finding the convex hull, finding of closest pair of points).

2-5 Most Important Required and Recommended Literature

List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

- J. Kleinberg, É. Tardos: Algorithm Design. Addison-Wesley, 2006.
- T.H. Cormen, C.E. Leiserson, R.L. Rivest, C Stein, Introduction to Algorithms. Fourth Edition. The MIT Press, 2022.

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.1
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

[1] **Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.

[2] e.g. case studies, role play, thematic presentations, etc.

[3] e.g. continuous debriefing, mid-year report

[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.

Course responsible (name, position, academic degree): **Dr. László Ferenc Szabó, Habil. Associate Professor, PhD**

Instructor(s) involved in teaching the course (if any) (name, position, academic degree):
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